

Crusher to the Bridge: Cruise Norovirus Calibration and Intervention Economics in an Agent-Based Shipboard Simulation

Definitive technical report for external review and distribution

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Modeling and technical report with embedded narrative review

Abstract

Crusher to the Bridge (CTB) is an open-source agent-based shipboard simulation framework used here as an enabling tool for cruise norovirus calibration and intervention economics. Cruise ships report norovirus outbreaks rarely despite frequent community norovirus PCR positivity among apparently healthy travelers. The modeling target was not community prevalence alone. It was the joint calibration of a low baseline reportable-outbreak rate of approximately 0.18–0.19% of voyages and an approximately 3.9-fold increase during GII.4 emergence periods, using CDC Vessel Sanitation Program (VSP) surveillance, norovirus shedding biology, partial immunity, and overdispersed transmission [1–5].

The calibrated branching-process component shows that these two targets constrain but do not uniquely identify the model. Plausible fits occur across combinations of $R_0 \approx 3.5$ –8.0, dispersion $k \approx 0.05$ –0.20, baseline susceptibility $s \approx 0.20$ –0.36, and $n_{\text{eff}} \approx 0.02$ –0.06 effective index cases per voyage. A rounded working parameterization, $R_0 \approx 4.5$, $k \approx 0.10$, $s \approx 0.30$, and $n_{\text{eff}} \approx 0.035$, reproduces baseline rarity; increasing susceptibility to approximately 0.65–0.70 reproduces emergence-period spikes without requiring a higher boarding prevalence. The interpretation is that cruise outbreak risk is governed by acute transmission-competent introductions entering a partially immune and highly overdispersed contact network, not by the total number of PCR-positive travelers.

Intervention economics follow from this calibration. Enhanced onboard response is favored under current cruise assumptions because it acts after rare introductions have declared themselves and has low incremental cost. Universal molecular pre-boarding screening is biologically coherent but economically inefficient at the baseline outbreak rate because most voyages contain no effective index case to remove. Pooled, gated, or seasonal screening can improve screening economics, but only under lower fully loaded assay costs, higher seasonal risk, or higher-consequence pathogen scenarios. CTB is designed for multi-context application; analyses of other vessel types will be presented separately. The software repository is available at [Crusher_to_the_Bridge](#).

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1 Introduction

Infectious-disease outbreaks on confined vessel platforms are a recurrent public-health and operational problem. Cruise ships are the best-documented case because the CDC VSP maintains a surveillance system for reportable acute gastroenteritis outbreaks, and because cruise ships concentrate large populations into shared dining, cabin, entertainment, excursion, and medical-care networks [1, 2, 6]. Similar modeling questions arise for expedition vessels, offshore platforms, and naval vessels, but cruise ships provide the most transparent public calibration data.

Randomized trials of shipboard outbreak prevention are rarely practical. The relevant interventions can change passenger behavior, crew workflows, diagnostic triage, isolation timing, cleaning intensity, food-service operations, and excursion policies. These interventions also interact with rare events and with known shipboard transmission routes, including person-to-person spread, shared cabins, food-service exposure, fomite contamination, and vomiting-associated environmental contamination [6–10]. If the base rate of a reportable outbreak is near one to several per thousand voyages, an adequately powered prospective trial of universal screening or wearable biosurveillance would be expensive, slow, and ethically difficult. Agent-based simulation is therefore useful as a decision-support tool: it allows counterfactual comparisons under matched random seeds, exposes sensitivity to uncertain biological parameters, and prevents high-regret operational concepts from being tested first on passengers or crew.

The motivating paradox for CTB is an “everything is everywhere” problem. Norovirus PCR positivity is not rare in the community. Community prevalence syntheses estimate a 7% global pooled prevalence of asymptomatic norovirus infection, with community estimates near 9% and lower values in Europe and North America [11]. Community incidence and attributable-disease studies show that a substantial fraction of norovirus infections are never captured by routine clinical surveillance [12–14]. Other community studies and shedding analyses show that PCR positivity can persist after symptoms resolve, often at lower titers [3, 4]. If even 1% of 3,000 boarding passengers are PCR-positive, dozens of infected travelers are expected at embarkation, yet reportable VSP outbreaks occur on only a small fraction of voyages. A useful shipboard biosurveillance model must resolve this discrepancy before it can rank interventions.

CTB was developed as an open-source, configuration-driven simulation platform using AI-assisted coding tools, including Devin and Cursor, bounded by SonarQube quality gates and external epidemiological packages. The repository is available at [Crusher_to_the_Bridge](#). The role of AI-assisted development is methodological: it accelerated implementation and refactoring while automated quality gates and simulation tests bounded software quality. The analysis presented here focuses on cruise ships because public outbreak data and cost assumptions are available. CTB is configurable for other confined platforms, but the cost-effectiveness results in this report apply to cruise ships unless otherwise stated.

2 Prior Work and Positioning

CTB is closest to the agent-based modeling (ABM) literature that represents individual mobility, contact structure, and intervention timing, but its cruise-norovirus analysis has a narrower calibration target. Kerkmann et al. model realistic human mobility and contacts to evaluate testing and isolation strategies; Hunter et al. use open data to construct a reusable infectious-disease ABM; Kou et al. couple within- and between-scale COVID-19 dynamics to assess vaccination and non-pharmaceutical interventions; Pan et al. apply ABM to cruise-ship spatial and operational design; and Wolpers et al. address intervention optimization for ABM simulations [15–19]. CTB differs by treating public VSP outbreak rarity and GII.4 emergence as calibration targets before comparing interventions.

Superspreading and overdispersion are also central. The CTB branching-process component uses a negative-binomial offspring distribution because heterogeneous infectiousness can dominate emergence probability. This follows work showing that superspreading changes intervention opportunities, that dispersion can be estimated from outbreak-size data, and that individual transmissibility can be decomposed into biologically and behaviorally distinct components [20–24]. In CTB, the decomposition is applied to norovirus: PCR positivity, acute high-titer shedding, behavioral non-exclusion at embarkation, susceptibility, and shipboard contact opportunity are separated rather than collapsed into a single prevalence term.

Table 1: Positioning CTB relative to recent ABM and superspreading-adjacent infectious-disease modeling. Validation status refers to the cited application, not to general software quality.

Model	Contact ture	struc- ture	Calibration target	Interventions	Validation status
Kerkmann et al. 2025 / arXiv 2410.08050	Realistic mobility and contact behavior	human and con-	Epidemic spread under mobility/contact assumptions	Testing and iso- lation strategies	Compared against mobility/contact reproduction targets and intervention scenarios
Hunter et al. 2018	Synthetic population from demographic and spatial data	popu- lation open and	Generic out- break dynamics in open-data setting	Disease-control scenarios config- urable by model user	Demonstrates reusable open-data ABM architecture
Kou et al. 2021	Multi-scale individual and population con- tacts	in- dividual and con-	COVID-19 epi- demic trajec- tories	Vaccination and non- pharmaceutical interventions	Scenario validation against COVID-19 case dynamics
Pan et al. 2026	Cruise-ship and operational contact patterns	spatial operational	Healthy cruise- ship design out- comes	Spatial and oper- ational design in- terventions	Cruise-design ABM application; not norovirus VSP calibration
Wolpers et al. 2026	ABM intervention search space	ABM intervention search space	Optimized intervention policies in infectious- disease simula- tions	Algorithmically optimized inter- ventions	Methods contribution for optimizing ABM interventions
CTB cruise norovirus analysis	Cabins, travel groups, passenger- crew work con- tacts, shared venues, environ- mental pathways	travel groups, passenger- crew work con- tacts, shared venues, environ- mental pathways	VSP baseline outbreak rarity and GII.4 emer- gence spike	Onboard re- sponse, univer- sal and gated screening, pool- ing, seasonal variants, wear- able triage	Calibrated to public VSP outbreak rates and norovirus biology; prospective embarkation-cohort validation remains absent

3 The Crusher_to_the_Bridge Simulation Platform

CTB is used here as an enabling simulation framework for cruise norovirus calibration and intervention economics. Agents represent passengers, crew, medical staff, food-service workers, housekeeping staff, and other functional groups. The cruise configuration assigns agents to cabins, travel groups, dining venues, work areas, medical spaces, shared circulation zones, and role-specific schedules. For norovirus, modeled transmission pathways include direct contact, food-service exposure, environmental contamination, fomite transfer, and vomiting-associated dispersal, reflecting evidence that cruise norovirus outbreaks are not attributable to a single route [6, 7, 10, 25–27].

The cost-effectiveness analysis uses a 5,000-passenger cruise scenario, while the larger software configuration includes crew and role structure. Population size, voyage duration, deck structure, cohort assignment, pathogen profile, diagnostic accuracy, alert thresholds, and response protocols are configuration inputs rather than hard-coded assumptions. The platform is designed for multi-context application; analyses of other vessel types will be presented separately. Supplementary Tables 6 and 7 list the configuration fields that must be replaced before CTB is applied outside the cruise norovirus setting.

CTB separates PCR positivity from transmission competence. Norovirus challenge and outbreak studies report high early fecal viral loads, prolonged PCR detectability, and lower titers during late shedding [3–5, 28]. The simulation therefore allows a traveler to be PCR-positive without being an effective index case for a reportable outbreak. This distinction is the bridge between the epidemiological calibration in Section 4 and the intervention economics in Sections 5–6.

Table 2: CTB diagnostic cascade. Sensitivity and specificity values are configurable scenario inputs, not universal device claims. Wearable or syndromic signals are treated as triage events, not diagnostic endpoints.

Tier	Layer	Evidence	Permitted response
T0	Wearable or syndromic triage	Physiological anomaly, self-report, or crew observation	Medical review; no fleet-level action from raw anomaly rates
T1	Symptom questionnaire and rapid test	Structured symptoms plus rapid diagnostic result	Preliminary isolation and targeted cleaning
T2	Molecular confirmation	PCR or equivalent molecular confirmation	Confirmed case management; escalation to ship medical leadership
T3	Operational response	Cluster confirmation and epidemiological review	Fleet-level protocols, cohorting, environmental response, and external reporting when criteria are met

4 Epidemiological Calibration

All simulation-derived values in this section are reproducible from the CTB repository ([Crusher_to_the_Bridge](#)) and from the branching-process equations, parameter ranges, and cost assumptions stated below. Internal execution identifiers have been removed; claims are tied either to peer-reviewed literature, public VSP surveillance, or the methods described in this report.

4.1 Community prevalence and the boarding problem

The boarding problem begins with prevalence, but prevalence is not the calibration endpoint. A pooled analysis of 71 studies estimated the pooled global prevalence of asymptomatic norovirus infection at 7%, with community settings near 9% and lower estimates in Europe and North America [11]. A Japanese cross-sectional PCR study of asymptomatic adults found 2.5% norovirus positivity, split between GI and GII genogroups [29]. Additional population and disease-attribution studies show that community norovirus incidence is large relative to the number of medically attended or reported cases, making direct inference from surveillance counts to infection counts unreliable [12–14]. Literature syntheses also support winter seasonality and prolonged shedding, meaning that a cruise voyage can plausibly board PCR-positive travelers who feel well at embarkation [3, 4, 28].

A simple boarding calculation illustrates the problem. At 1% prevalence, a 3,000-passenger ship expects about 30 PCR-positive passengers. A Monte Carlo boarding model that accounted for clustering estimated approximately 13 infected people in about six independent clusters under one 1% scenario, and the probability of at least three infected boarders exceeded 97%. Figure 1 shows the distribution expected from independent binomial boarding under several prevalence assumptions. The figure is deliberately simple: it shows why PCR positivity cannot be mapped directly to outbreak risk; later calibration uses ranges for transmission competence, overdispersion, and susceptibility rather than a single prevalence-to-outbreak conversion.

The calibration therefore separates three states that are often conflated. The first state is molecular positivity: a stool or rectal sample contains detectable viral RNA. The second state is biological infectiousness: the person is shedding virus at a level and in a phase where onward transmission is plausible. The third state is operational effectiveness: the person boards, mixes with susceptible contacts, contaminates relevant environments, and is not isolated quickly enough for stochastic extinction to occur. Challenge and outbreak studies show that RNA shedding can persist for weeks, while symptomatic cases, asymptomatic infections, and convalescent shedders can have different viral-load distributions and different opportunities for exposure [3, 4, 28, 30]. Ge et al. further support an inoculum-dose relationship with early virus kinetics and symptoms, which is relevant to the assumption that acute high-titer shedders are the most detectable and most operationally important group, although dose effects on total shedding were not definitive [31]. The effective-index variable in CTB is designed to represent the third state, not the count of PCR-positive travelers.

4.2 Observed outbreak rates

CDC VSP data define the cruise calibration target. The VSP outbreak line list and MIDRS-derived calibration analysis contained 424 outbreaks over 1993–2026. Norovirus accounted for approximately 86% of confirmed outbreaks. Overall rates were approximately 4.2 outbreaks per 1,000 voyages across the long series and declined to about 1.3 per 1,000 voyages in 2014–2019, depending on denominator assumptions. Published cruise surveillance studies provide the external anchor: VSP analyses found that norovirus dominates confirmed cruise acute gastroenteritis outbreaks, while earlier work showed outbreak increases despite

Community prevalence predicts frequent PCR-positive boarding

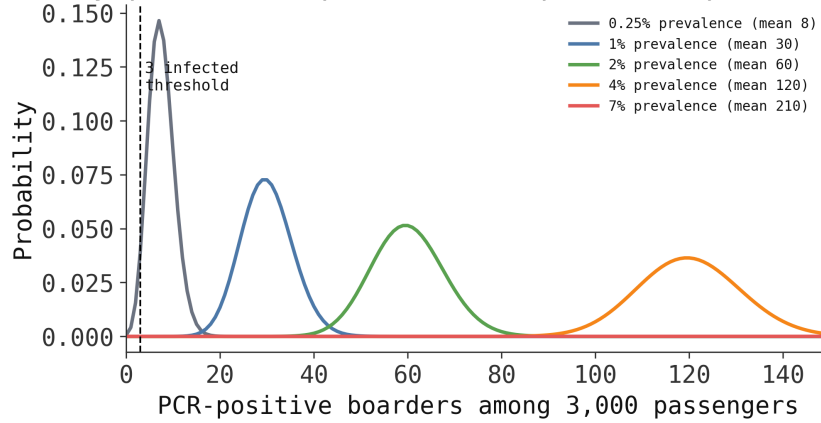


Figure 1: Distribution of PCR-positive boarders among 3,000 passengers under alternative community prevalence assumptions. At 1% prevalence the expected count is 30, and the probability of at least three PCR-positive boarders is effectively one. The figure is a boarding arithmetic visualization generated from the prevalence scenarios used in CTB calibration.

sanitation inspection programs, supporting a model in which person-to-person, cabin, food-service, and environmental pathways remain important [1, 6–9]. The VSP reporting threshold means these are reportable outbreaks rather than all introductions or all small transmission chains.

Norovirus passenger attack rates in the calibration data had a median of 5.6% and were well approximated by a lognormal distribution with $\mu = 1.80$ and $\sigma = 0.51$ on the percentage scale. Outbreak timing showed strong seasonality: approximately 43% of outbreaks occurred in January–March. Cruise-specific systematic review evidence is consistent with this calibration target. Mouchtouri et al. reported a weighted passenger attack rate near 7%, a lower crew attack rate near 2%, person-to-person spread in most outbreaks, and a large cabin-mate association, which supports cabin and travel-group clustering as a modeled risk amplifier rather than an explanation for all outbreak rarity [2]. Figure 2 summarizes the VSP outbreak calibration plots used to set these targets.

4.3 The approximately 400-fold paradox and its resolution

If dozens of PCR-positive travelers board and if CTB simulations with three index cases produce outbreaks in nearly every high-severity run, cruise outbreaks should be common. They are not. The reconciliation requires three filters that act before an introduction becomes a reportable outbreak.

First, only a small fraction of PCR-positive boarders are effective index cases. The effective-index analysis modeled a shedding curve with high early titers and prolonged low-titer PCR detectability. Among PCR-positive shedders at an arbitrary point in time, 12.3% were in an acute high-titer phase above 10^8 copies/g, 11.0% were in a declining 10^6 – 10^8 copies/g phase, and 76.6% were low-titer shedders below 10^6 copies/g. Behavioral filters further reduce effective boarding because symptomatic travelers may self-exclude or be detected by routine health screening. Symptomatic cases appear to contribute disproportionately in some institutional settings, while asymptomatic family members can still mediate household transmission; this mixed evidence supports a graded transmission-competence model rather than a binary symptomatic-only assumption [4, 30, 32]. The fitted cruise model required only about 0.03% of PCR-positive boarders to function as transmission-competent effective index cases.

Second, transmission is overdispersed. A negative-binomial offspring distribution with dispersion near $k = 0.10$ means that most introductions infect nobody, while a small fraction generate many secondary infections. This is consistent with shipboard and household evidence that close contacts, food pathways, and environmental contamination can create clustered transmission rather than smooth homogeneous spread [7, 8, 33]. The use of a negative-binomial offspring model follows the infectious-disease superspreading literature, in which individual variation strongly changes emergence probability, and the dispersion parameter k is estimated with methods developed for highly overdispersed outbreak data [23, 24, 34]. In the CTB reconciliation, overdispersion contributed an additional approximately 5–10-fold extinction filter relative to a non-overdispersed outbreak model.

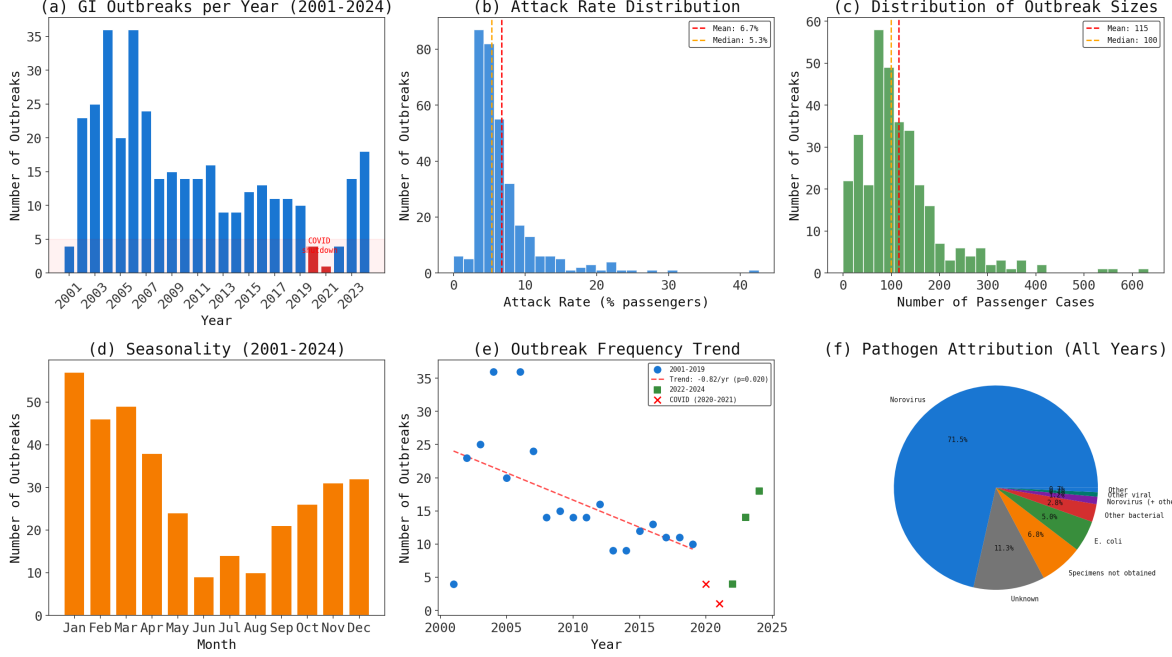


Figure 2: CDC VSP outbreak calibration plots used in CTB model fitting. The data establish norovirus dominance among confirmed cruise gastroenteritis outbreaks, the empirical passenger attack-rate distribution, and seasonal concentration of reportable outbreaks.

Third, population immunity reduces susceptibility. Norovirus immunity is incomplete, genotype-specific, and variable across hosts, but adult populations are not immunologically naive. Challenge and serological studies support partial protection after exposure and measurable correlates of protection against GII.4 disease [35–37]. Birth-cohort and immunological reviews also support repeated infection, genotype-specific protection, and host-genetic susceptibility as reasons why the susceptible fraction is neither zero nor one [38, 39]. The reconciliation model fit non-emergence periods with a susceptible fraction near 0.30 and GII.4 emergence periods with susceptibility rising to approximately 0.65–0.70.

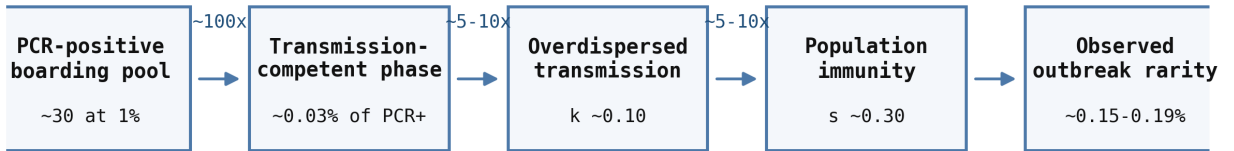


Figure 3: Mechanistic decomposition of the prevalence-versus-rarity paradox. PCR-positive boarding counts are high, but effective outbreaks are filtered by the rare acute transmission-competent shedding phase, overdispersed extinction, and partial population immunity. Plausible ranges for R_0 , k , susceptibility, and n_{eff} are reported in Tables 3 and 4.

4.4 The reconciliation model

The reconciliation model is a branching-process calculation used to fit simulation inputs to two external constraints: rare baseline VSP outbreaks and a higher outbreak rate during GII.4 emergence. It is not a direct measurement of the number of infectious passengers. Let R_0 denote the pathogen reproduction number in a fully susceptible shipboard population, let s denote the susceptible fraction, and let $R_{\text{eff}} = sR_0$ denote the reproduction number after immunity is applied. Offspring counts follow a negative-binomial distribution

with dispersion k . For one effective index case, the extinction probability q solves

$$q = \left(\frac{k}{k + R_{\text{eff}}(1 - q)} \right)^k.$$

For n effective index cases, the outbreak probability is $1 - q^n$. The model then searches parameter combinations that match the observed cruise baseline and emergence spike while remaining biologically plausible. This likelihood-style use of k follows the overdispersion estimation framework for outbreak data, but the CTB values remain calibrated parameters rather than externally measured constants [23, 24, 34].

The best practical parameter set for simulation used $R_0 \approx 4.5$ (plausible range 3.5–8.0), dispersion $k = 0.10$ (plausible range 0.05–0.20), baseline susceptibility ≈ 0.30 (range 0.20–0.36), and $n_{\text{eff}} = 0.035$ (plausible range 0.02–0.05) effective index cases per voyage. These are fitted or estimated calibration parameters, not directly measured values from a cruise cohort. This parameterization predicts a baseline outbreak rate near 0.15%, close to the 0.181–0.19% baseline rate in the structured reconciliation data. A more exhaustive grid identified a closely related best match with $R_0 = 4.5$, $k = 0.05$, baseline susceptibility 0.29, emergence susceptibility 0.70, and $n_{\text{eff}} = 0.06$, predicting a 0.146% baseline rate and a 0.57% emergence rate. The recommended simulation values intentionally use rounded, interpretable parameters rather than overfitting a single grid point.

Table 3: Calibration parameters used to reconcile community PCR positivity with rare reportable cruise outbreaks. Values are simulation estimates fitted to VSP outbreak rarity and norovirus biology, not direct measurements.

Parameter	Working value	Interpretation and uncertainty
R_0	≈ 4.5	Plausible range 3.5–8.0. Represents transmission potential in a fully susceptible shipboard population.
k	0.10	Plausible range 0.05–0.20. Lower values imply stronger overdispersion and more stochastic extinction.
Baseline susceptibility	≈ 0.30	Range 0.20–0.36. Represents the effective susceptible fraction after partial immunity and host heterogeneity.
n_{eff}	0.035 per voyage	Plausible range 0.02–0.05. Represents acute, transmission-competent, behaviorally non-excluded effective index cases, not PCR-positive boarders.
Emergence susceptibility	0.65–0.70	Fitted high-susceptibility state used to reproduce GIL4 emergence spikes.

4.4.1 Parameter identifiability and sensitivity

The two calibration targets, a baseline reportable-outbreak rate near 0.18–0.19% of voyages and an approximately 3.9-fold GIL4 emergence spike, constrain but do not uniquely determine the four-parameter model ($R_0, k, s, n_{\text{eff}}$). This is a structural identifiability problem: changes in R_0 can be partly offset by changes in susceptibility, dispersion, or the effective-index rate, and a single pair of outbreak-rate targets cannot identify all four quantities without prior biological constraints. The sensitivity analysis therefore reports a parameter region rather than a single fitted truth, following the principle that epidemic model forecasts should propagate parameter uncertainty rather than report one optimum [40].

Table 4: Examples of parameter combinations that reproduce the baseline rarity and emergence-spike constraints. Outbreak probabilities are branching-process predictions per voyage. The table illustrates non-identifiability; it is not a ranking of biologically preferred parameter sets.

R_0	k	Baseline s	Emergence s	n_{eff}	Baseline outbreak %	Emergence fold-change
7.0	0.20	0.20	0.60	0.020	0.216	3.88
7.0	0.15	0.20	0.60	0.020	0.169	3.85
7.0	0.05	0.20	0.65	0.060	0.183	3.98
7.0	0.10	0.20	0.60	0.030	0.176	3.82
3.5	0.20	0.35	0.65	0.030	0.198	3.81
3.5	0.15	0.35	0.65	0.035	0.181	3.79
4.5	0.10	0.30	0.68	0.035	0.184	3.42
4.5	0.05	0.29	0.70	0.060	0.146	3.81

The predictions most sensitive to this uncertainty are interventions that act before embarkation. Screening value scales with n_{eff} , assay sensitivity among acute high-titer shedders, and the probability that removing

one effective index case prevents a reportable outbreak. Onboard response is less sensitive to the prevalence side of the calibration because it acts after symptomatic or clustered cases have appeared, but its predicted case reduction remains sensitive to attack-rate distributions, isolation delay, and compliance.

GII.4 emergence provides an external validation. New GII.4 variants have historically altered norovirus epidemic dynamics through antigenic change and population immune escape [41–44]. Recent cruise genotype surveillance also supports continued genotype monitoring as part of outbreak interpretation, although the Preston et al. report is an Open Forum Infectious Diseases abstract and should be treated as surveillance evidence rather than a full mechanistic study [45]. In CTB reconciliation, increasing susceptibility from approximately 0.30 to 0.65–0.70 raised R_{eff} to approximately 2.9–3.15 and reproduced the observed 3.9-fold outbreak spike during GII.4 emergence without increasing the effective index count. This matters because it identifies strain novelty, not passenger prevalence alone, as the parameter most capable of producing temporal clusters of cruise outbreaks.

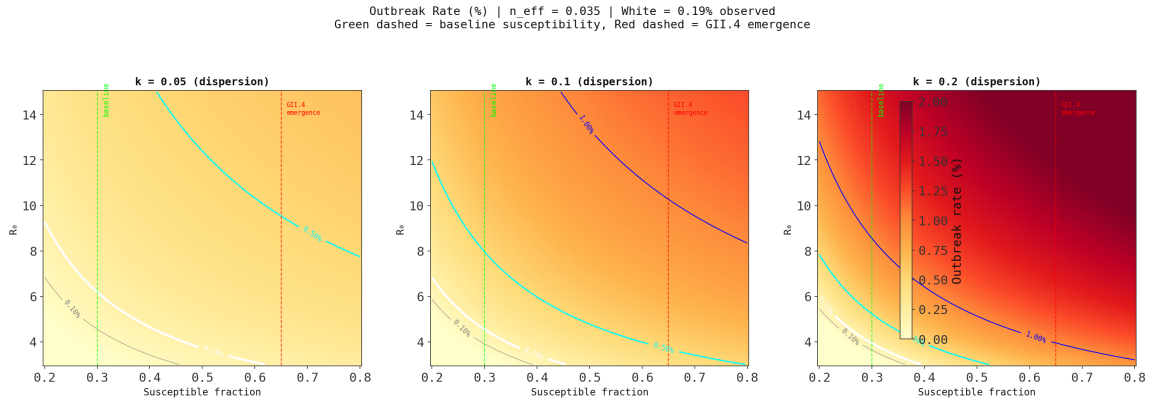


Figure 4: Sensitivity of outbreak probability to true R_0 and susceptible fraction. The calibrated baseline lies near $R_0 = 4.5$ and susceptibility 0.30, while GII.4 emergence is represented primarily as an increase in susceptibility. The contour structure shows why multiple (R_0, s) pairs can produce similar outbreak rates.

4.5 Household clustering and the travel-group effect

Travel groups create correlation among boarders. Household secondary attack rates for norovirus are commonly near 20%, with reported ranges spanning approximately 13–48% depending on setting, genotype, ascertainment, and study design [33, 34, 46]. Cruise investigations identify cabin sharing as one of the strongest risk factors; Mouchtouri et al. report a cabin-mate odds ratio of 38.7 in the cruise outbreak literature [2]. A serial interval near three days means secondary cabin or household cases can develop before disembarkation on a one-week cruise. Travel-group clustering therefore explains why one true introduction can board with or generate several linked infections, but it does not eliminate the need for the effective-index filter.

The distinction matters for screening. A cabin mate of a symptomatic case is highly enriched for exposure, but a random embarkation passenger is not. CTB therefore treats group structure as a way to create correlated positives and rapid early clusters, not as evidence that universal screening is cost-effective. In practice, travel-group information is more useful after a signal appears: cabin mates and dining companions can be prioritized for testing, symptom checks, and isolation support. That operational approach aligns better with the observed cabin effect and with household evidence for asymptomatic transmission than a model that assumes all PCR-positive travelers pose equal risk [2, 32, 33].

5 Interventions Explored

Table 5 consolidates the intervention portfolio. Simulation-derived entries use the cruise norovirus calibration in Section 4; literature-anchored entries are biologically motivated scenarios whose economics depend on operational gating, assay cost, and seasonal deployment.

Table 5: Consolidated intervention comparison for cruise norovirus. Costs are per voyage unless noted. Reportable-outbreak effects are relative to the standard-response simulation when a direct simulation estimate is available.

Strategy	Mechanism	Effect on outbreaks	Effect on cases	Cost/voyage	Cost/outbreak prevented	Evidence strength
Standard response	Baseline hygiene, routine case management	Baseline: 65 reportable outbreaks per 10,000 simulated voyages	Baseline: 23,332 cases per 10,000 voyages	\$10,177 mean all-in	Reference	Literature-anchored VSP calibration plus simulation baseline
Enhanced on-board	Earlier detection, isolation, cleaning, cabin/contact management	53 reportable outbreaks per 10,000 voyages; 12 prevented	7,373 cases averted per 10,000 voyages	\$10,178 mean all-in; \$2 incremental	\$1,400 per reportable outbreak prevented	Simulation-derived; mechanism literature-anchored
Universal PCR	Molecular screening of all passengers before boarding	5 reportable outbreaks per 10,000 voyages; 60 prevented	20,802 cases averted per 10,000 voyages	\$280,442 mean all-in	\$45.0M per reportable outbreak prevented	Simulation-derived; assay performance literature-anchored
Questionnaire-gated	Symptom/travel-group gate followed by molecular or rapid testing	Expected to reduce test volume but misses asymptomatic acute shedders	Case effect depends on gate sensitivity and compliance	Not fixed; proportional to gate-positive fraction	Not estimated as a robust point value	Literature-anchored scenario; requires prospective gate validation
Thermal-gated	Temperature gate followed by testing or denial of boarding	Limited for norovirus because fever is not a reliable acute-enteric screen	Expected case reduction low unless paired with symptoms/testing	Low direct cost; high false-negative risk	Not estimated as a robust point value	Literature-anchored weak evidence for norovirus
Warm-towel pooled	Low-friction pooled molecular sampling at embarkation	Can improve economics if fully loaded test costs are near sub-dollar breakeven	Case effect depends on acute-shedder sensitivity and pool dilution	Matrix pooling: 491 tests for 5,000 passengers at 32x32 design	Breakeven near \$0.73 per test in cruise base case	Simulation-derived pooling arithmetic; no direct cruise-norovirus validation
Seasonal variants	Deploy screening during winter or GII.4 emergence periods	Higher value when susceptibility or effective-index rate rises	Higher value if deployed during high-risk periods	Annualized cost depends on duty cycle	Not a single value; threshold set by seasonal risk multiplier	Simulation-derived sensitivity; seasonality and genotype biology literature-anchored

5.1 Wearable biosurveillance and the crew-first strategy

The earlier CTB wearable report tested a diagnostic cascade that begins with wearable physiological anomaly detection and proceeds through confounder-aware triage, symptom validation, rapid testing, and molecular confirmation. The report identified the central safety problem: wearables generate signals in healthy people. In the maritime setting, seasickness, exertion, alcohol, poor sleep, heat stress, medication, heat exposure, and schedule disruption can all resemble infection in consumer physiological streams. External evidence supports wearable detection of SARS-CoV-2 and other febrile or respiratory infections, but it does not establish pre-symptomatic detection of enteric pathogens such as norovirus. Mitratza et al. reviewed wearable performance for SARS-CoV-2 detection, and Ming et al. described continuous physiological monitoring for infectious-disease management and outbreak response; both are best interpreted here as respiratory-pathogen and general biosurveillance analogies rather than direct norovirus validation [47, 48].

The cascade is therefore a risk-control device as much as a detection system. Raw wearable anomalies enter a low-regret triage state. Escalation requires symptom review, contextual filtering, and molecular confirmation before ship-level operations change. This architecture avoids treating resting heart rate, heart-rate variability, skin temperature, or motion-derived sleep changes as a diagnosis. It also makes the system auditable: each action can be traced to a tier rather than to an uncalibrated anomaly count. The earlier report found that naive escalation from raw anomalies can increase infections by forcing contacts into higher-risk workflows and by creating avoidable crew movement during lockdown-like responses. This is consistent with the broader lesson from confined-setting outbreaks: operational response can change contact networks and therefore become part of the transmission system rather than an external intervention [49].

The corrected wearable deployment comparison found that crew-only monitoring captured most of the modeled surveillance benefit. In the corrected runs, crew-only monitoring involved 171 monitored agents, crew plus passenger bring-your-own-device monitoring involved 292 monitored agents, and a more incentivized passenger-inclusive arm involved 352 monitored agents. Passenger monitoring did not add a statistically significant benefit beyond crew-only monitoring in the corrected comparison: Crew-to-BYOD $p = 0.63$ for SARS-CoV-2 and $p = 0.43$ for norovirus. Across eight pathogen classes, respiratory pathogens benefited more than norovirus: influenza A infections fell by 65% ($p < 0.0001$), SARS-CoV-2 by 54% ($p = 0.014$), and norovirus GII.4 showed a 20% positive trend ($p = 0.163$). In a 100-cruise norovirus power run, mean infections decreased from 130.8 to 120.4, an 8.0% reduction ($p = 0.050$, $d = 0.20$).

The crew-first result has a practical basis. Crew traverse more ship zones than individual passengers, interface with dining and housekeeping workflows, and remain on board across repeated passenger cohorts. A crew sentinel system also avoids many passenger consent, privacy, device-support, and attrition problems. The limitation is biological: norovirus can move through vomiting, shared cabins, food-service contamination, and fomites with a short incubation period, so earlier physiological detection may not precede meaningful environmental spread. There is no published empirical evidence that consumer wearables can detect norovirus or other enteric-pathogen infections pre-symptomatically with operationally useful sensitivity and specificity. The CTB result should therefore be read as a simulation of a guarded surveillance architecture under assumed sensor performance, not as validation that wearables can prevent cruise norovirus outbreaks.

These findings place wearable biosurveillance in the intervention portfolio but do not make it the preferred norovirus prevention strategy. For norovirus, the cascade is useful mainly as a guarded detection and triage architecture. The demonstrated prevention effect is limited by the short incubation period, rapid environmental contamination, and seed-to-seed stochasticity. It is therefore less economically decisive than enhanced onboard case finding and isolation in the cruise norovirus scenario. For respiratory pathogens, where host physiological changes and pre-symptomatic shedding may create a larger detection window, the same crew-first architecture may be more valuable, but those results should not be imported into enteric-pathogen policy without pathogen-specific validation [47, 48, 50, 51].

5.2 Enhanced onboard response

The enhanced onboard response scenario combined active surveillance, 50% case detection, and 72-hour isolation. It reduced reportable outbreaks from 65 to 53 per 10,000 simulated voyages and reduced cases from 23,332 to 15,959 per 10,000 voyages. The incremental mean cost was about \$2 per voyage relative to standard response in the cost model, yielding approximately \$2 per case averted and about \$1,400 per reportable outbreak prevented.

This intervention is damage control rather than introduction prevention. In the simulation it did not reduce the number of outbreak initiations. Instead, it shortened chains, reduced final size, and lowered the probability that an introduced chain crossed the reportable threshold. That distinction is operationally important.

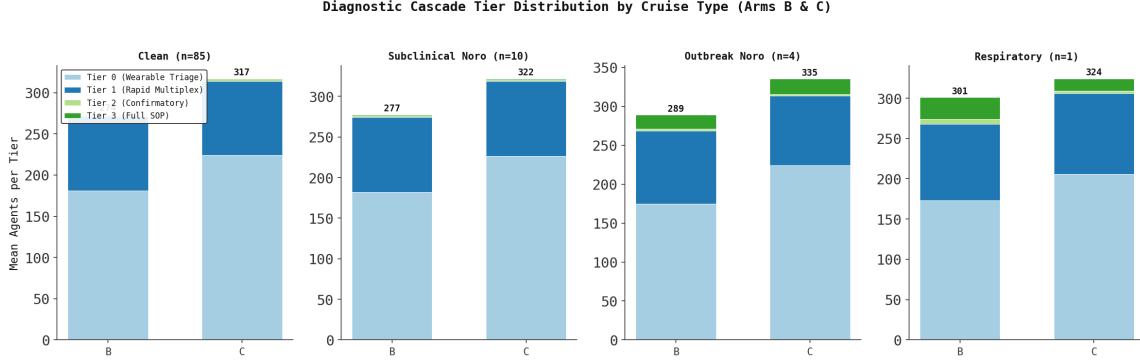


Figure 5: Diagnostic cascade tier distribution from the wearable biosurveillance analysis. The cascade filters numerous early signals before confirmation and prevents raw anomaly counts from directly triggering operational escalation.

Enhanced response is economically attractive because it is deployed after evidence appears and because isolation is targeted to cases. It cannot remove infected boarders before embarkation. The biological rationale is supported by evidence that symptomatic cases can be important transmission sources and that rapid case management, cleaning, and food-handler exclusion target the highest-risk interval [10, 27, 30, 52].

5.3 Universal pre-boarding screening

Universal pre-boarding screening is mechanistically sound. If a molecular or antigen test removes the rare effective index case before boarding, the downstream outbreak chain is prevented. The cost-effectiveness problem is the base rate. The calibrated model estimates $n_{\text{eff}} = 0.035$ (plausible range 0.02–0.05) per voyage, so approximately 96.6% of voyages have zero effective index cases under a Poisson approximation. Universal screening therefore spends most of its budget on voyages in which there is no transmission-competent boarder to intercept.

At \$55 per PCR test for a 5,000-passenger voyage, universal molecular screening costs about \$275,000 before any response costs. The cost-effectiveness model estimated mean per-voyage cost near \$280,442 for PCR screening, compared with about \$10,177–\$10,178 for standard or enhanced onboard response. Universal PCR screening reduced simulated reportable outbreaks from 65 to 5 per 10,000 voyages and cases from 23,332 to 2,530, but the incremental cost implied tens of millions of dollars per outbreak prevented. The structured cost task reported PCR breakeven only above approximately \$65 million per prevented outbreak, and the user-specified brief summarized the same result as approximately \$55 million per outbreak prevented under related assumptions. Universal rapid diagnostic testing at \$10 per passenger costs about \$50,000 per voyage and has lower sensitivity, leaving it economically unattractive in the cruise base-rate setting.

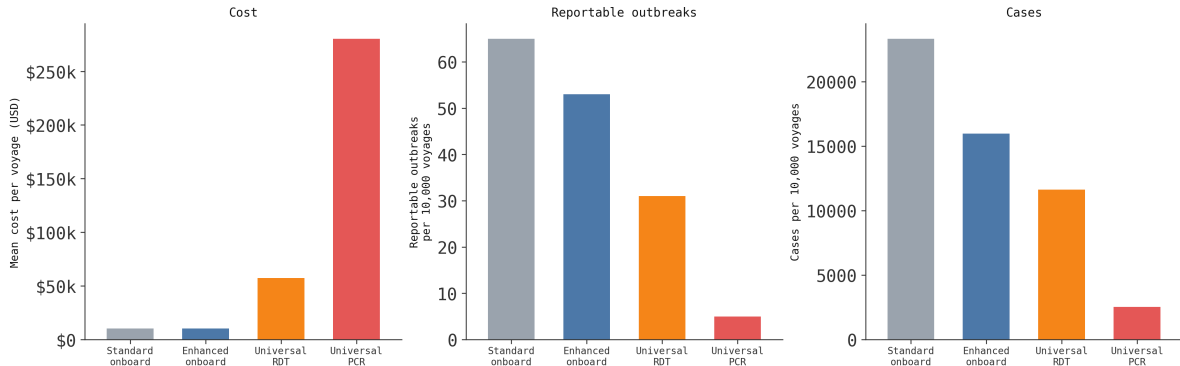


Figure 6: Cruise norovirus intervention comparison from the structured cost-effectiveness model. Universal screening lowers cases and reportable outbreaks but has high fixed cost per voyage. Enhanced onboard response is economically dominant because it is targeted after cases appear.

5.4 Simulation pre-boarding screening results

Matched-seed CTB pre-boarding screening experiments compared one, two, and three index-case conditions. Outbreak probability increased from 30% with one index case to 60% with two index cases and 100% with three index cases. The mechanism was stochastic extinction: screening prevented outbreaks by removing index cases and allowing chains to go extinct, not by reducing final outbreak size after a chain had already established. In the high-index scenarios, simulated attack rates of approximately 12–17% exceeded the empirical VSP median of 5.6%, so these runs represent a high-severity or worst-case regime rather than the typical reportable outbreak.

6 Viable Screening Strategies and Cost-Effectiveness

6.1 Gated screening approaches

Gated screening improves the economics by testing a smaller, enriched set of travelers. The cost-effectiveness analysis evaluated questionnaire-gated rapid testing at approximately \$3,200 per voyage and thermal-gated swabbing at approximately \$4,250 per voyage. The estimated breakeven reputational cost was \$321,000 for questionnaire-gated rapid testing and \$469,000 for thermal-gated swabbing. These values are closer to plausible cruise outbreak costs than universal screening, but they still depend on enrichment, compliance, staffing, and the ability to identify the rare transmission-competent shedder before boarding.

Seasonal deployment changes the base rate. Because approximately 43% of VSP outbreaks occur in January–March, restricting higher-cost screening to the winter quarter improves economics by roughly twofold relative to year-round deployment under the same assumptions. The practical implementation concept is a hospitality-compatible collection event, such as a warm hand towel or embarkation hygiene gesture that doubles as molecular sampling. The value of such a design is not the towel. It is that sample collection can be integrated into a routine that passengers already accept.

Gating also reduces harms from false positives. A positive norovirus molecular test in a low-risk, asymptomatic traveler can represent residual shedding rather than an acute high-risk state. A triage design that combines symptoms, known exposure, travel-group information, voyage season, and local outbreak context is more consistent with the effective-index model than universal action on PCR positivity alone [3, 4, 53]. For this reason CTB treats screening as a configurable risk-enrichment step rather than a fixed mandate.

6.2 Pooled testing analysis

The pooled testing review found no direct empirical literature validating pooled norovirus PCR or LAMP for cruise embarkation screening. This is an evidence gap and changes how the results should be interpreted. The CTB pooled-testing section is a modeling exercise that extrapolates from molecular pooling studies, especially SARS-CoV-2 work, not an established norovirus testing protocol. Published SARS-CoV-2 pooling studies support the feasibility of pooling under low prevalence, quantify sensitivity loss from dilution, and compare hierarchical or combinatorial designs in settings where individual testing would be expensive [54–56]. Cost analyses of SARS-CoV-2 testing similarly show that pooling economics depend on prevalence, pool size, labor, retesting burden, and the fully loaded cost per assay [57, 58]. These studies justify the mathematical extrapolation but do not validate pooled norovirus PCR/LAMP in stool, vomitus, environmental swabs, or embarkation collection workflows.

The CTB model used a dilution approximation in which cycle threshold (Ct) shifts by approximately one cycle per doubling of pool size. It treated acute transmission-competent shedders as high-titer samples with individual Ct values in ranges where dilution remains detectable through large pools. Under these assumptions, transmission-competent shedders with Ct 15–28 maintained 100% modeled sensitivity through pool size 64, while low-titer non-transmission-competent shedders were intentionally less reliably detected. The interpretation is conditional: if acute high-titer shedders dominate outbreak risk, then pooling can be designed to preserve detection of the subgroup that matters most operationally. If lower-titer or intermittently shedding individuals contribute more than assumed, the modeled sensitivity would be too optimistic.

Dorfman pooling and matrix pooling solve different operating problems. Dorfman pooling uses a first-stage pool followed by individual retesting of positive pools. It is simple but inefficient when a ship has thousands of travelers and low expected positivity. Matrix pooling places samples in rows and columns, allowing positives to be identified by intersections with fewer expected assays. For the cruise scenario with 5,000 passengers, the best matrix operating points used 16–64-dimensional designs and expected approximately 491–701 molecular assays for the voyage, depending on matrix size and positivity assumptions. Matrix designs also create operational risks not captured by the assay-count metric: sample tracking errors, repeat collection,

delayed positive resolution, and laboratory workflow failures can eliminate the modeled advantage. A cruise implementation would need empirical validation with the actual sample type, assay chemistry, collection device, sample transport time, and retesting workflow before deployment.

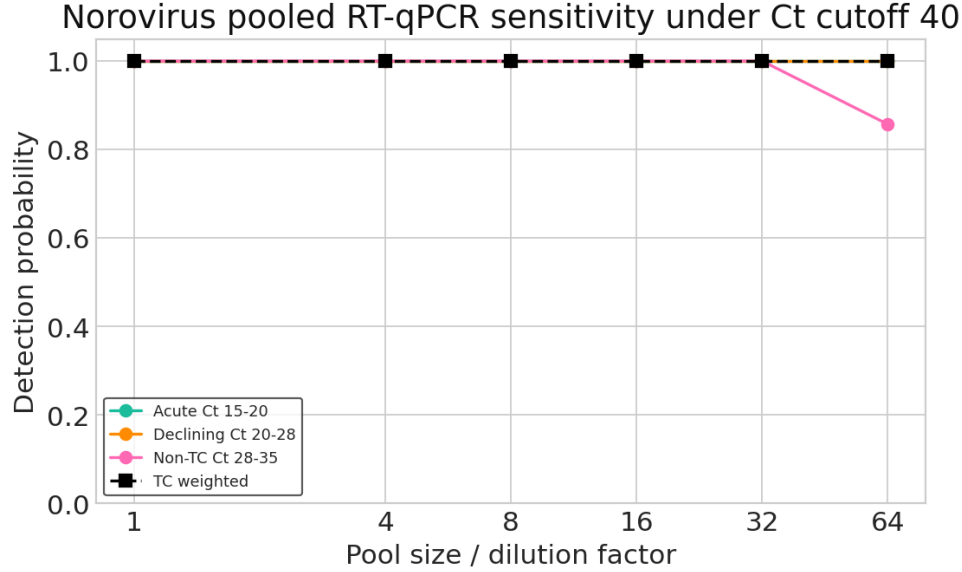


Figure 7: Modeled sensitivity of pooled molecular testing as a function of pool size and shedding phase. The model extrapolates from molecular pooling literature because direct pooled norovirus PCR/LAMP validation data were not found.

6.3 Future test cost trajectories

The cost trajectory for molecular diagnostics is the main reason pooled screening remains relevant. Current RT-LAMP assays are often modeled or reported in the approximately \$3–\$8 per-test range, depending on whether the estimate is reagent-only or fully loaded. CRISPR-based approaches such as SHERLOCK and DETECTR are approaching approximately \$1–\$3 per test in some projections and implementations, while paper-based and microfluidic approaches are sometimes projected below \$1 per assay over a five- to ten-year horizon [59–61].

For cruise ships, the modeled breakeven cost for matrix-pooled molecular screening was approximately \$0.51–\$0.73 per assay across the best matrix designs. The 32-dimensional matrix design required about 491 assays and had a breakeven cost of \$0.73 per test; the 64-dimensional matrix design required about 592 assays and had a breakeven cost of \$0.60 per test. These values imply that pooled cruise screening becomes economically plausible only when the fully loaded cost per molecular assay approaches roughly \$0.50–\$0.75, or when outbreak costs or baseline risk are much higher than the calibrated cruise estimate. SARS-CoV-2 cost-effectiveness studies are useful comparators because they show how prevalence and consequence severity can move the optimum, but their policy conclusions should not be transferred to norovirus without adjusting base rate, isolation cost, disease severity, and sample logistics [57, 58].

6.4 High-consequence pathogen screening

High-consequence pathogens change the decision calculus because the loss function is not dominated by common low-fatality gastroenteritis. Andes hantavirus is the relevant maritime example because person-to-person transmission has been documented for Andes virus, and exposure can occur during excursions in endemic regions [62, 63]. A reported MV *Hondius* outbreak in 2026 followed a birdwatching excursion and involved 11 cases, 3 deaths, and passengers returning to 23 countries [64]. For such hazards, pre-excursion risk assessment, excursion exclusion rules, rodent-exposure avoidance, and targeted post-exposure monitoring may dominate universal passenger screening. The intervention target is the high-risk exposure event, not the average embarkation passenger.

Matrix-pooled screening requires sub-dollar tests

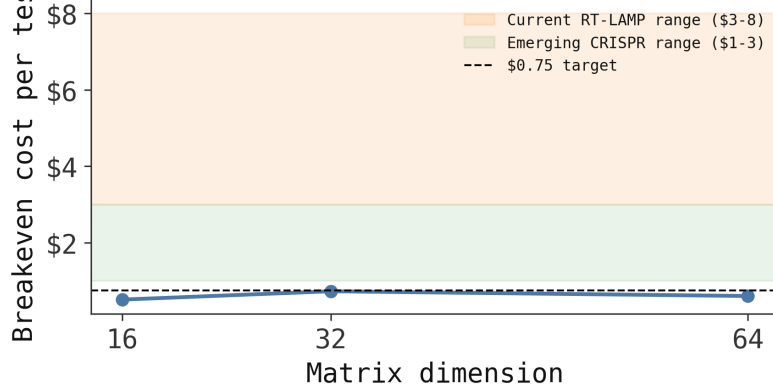


Figure 8: Cruise matrix-pooled molecular screening breakeven costs. The shaded bands show approximate current and emerging test-cost ranges from the diagnostic cost literature. Under calibrated cruise norovirus base rates, economically viable pooled screening requires sub-dollar fully loaded molecular assays.

7 Discussion

The CTB analysis supports the “everything is everywhere” principle but limits its operational meaning. PCR-positive pathogens can be common among apparently healthy travelers, yet outbreaks remain rare because PCR positivity is not equivalent to an effective introduction. For cruise norovirus, the decisive state is the rare conjunction of an acute high-titer shedder, non-exclusion at boarding, sufficient susceptibility, and a contact or environmental pathway that survives stochastic extinction. This conclusion is consistent with prolonged shedding and asymptomatic-infection literature, but it depends on fitted filters because no cruise embarkation cohort has directly measured all relevant states at once [3, 4, 11, 28, 31].

This interpretation explains why many interventions appear disappointing. Universal screening acts before boarding and can prevent introductions, but the number needed to screen is high because most voyages contain no effective index case. Wearable biosurveillance can move detection earlier for some pathogens, but norovirus has a short pre-symptomatic window and strong environmental pathways, so earlier detection does not always translate into fewer cases. Enhanced onboard response does not prevent introduction, but it is economically efficient because it spends resources after the signal is enriched. This is a base-rate result, not a claim that screening lacks biological value.

The super-shedder bottleneck is the operational target. Screening all passengers is expensive because the target is rare. A more plausible strategy is to enrich the target through season, itinerary, exposure history, cabin linkage, symptom reports, crew sentinel signals, and local outbreak intelligence. That approach is also more consistent with the cabin-mate odds ratio in cruise outbreaks and with household transmission studies showing clustering among close contacts [2, 32, 33, 46]. The practical implication is that risk enrichment and rapid onboard response should be evaluated together rather than as competing concepts.

Several limitations follow directly from the evidence base. First, the reconciliation parameters are fitted estimates with uncertainty: $n_{\text{eff}} = 0.035$ (plausible range 0.02–0.05), $k = 0.10$ (plausible range 0.05–0.20), baseline susceptibility ≈ 0.30 (range 0.20–0.36), and $R_0 \approx 4.5$ (plausible range 3.5–8.0). They should not be quoted as measured cruise values. Second, VSP data are reportable-outbreak data, not introduction data. Subthreshold chains and unreported mild illness are not fully observed. Third, the model uses literature-derived shedding and immunity assumptions from challenge studies, community cohorts, outbreak investigations, and healthcare settings that do not perfectly match cruise embarkation. Fourth, intervention cost estimates are cruise-specific and omit military/naval cost speculation by design.

Pooling has a specific limitation: no published empirical validation was found for pooled norovirus PCR or LAMP as an embarkation screen. The pooling analysis relies on dilution and assay-count mathematics extrapolated from SARS-CoV-2 and other molecular pooling studies [54–58]. Before operational use, the model would need validation against the actual cruise sample type and collection workflow. The relevant validation endpoint is not only analytical sensitivity in a laboratory pool. It is end-to-end detection of acute transmission-competent shedders fast enough to change boarding or early-voyage operations.

Wearable biosurveillance has a parallel limitation. The external wearable evidence base is from SARS-CoV-2, respiratory infections, and general infectious-disease monitoring, not norovirus-specific pre-symptomatic detection [47, 48]. The CTB wearable results therefore support a guarded diagnostic cascade and a crew-first deployment architecture under assumed signal behavior. They do not prove that wearables can identify enteric-pathogen infections before symptoms. Future work should test whether gastrointestinal illness produces a sufficiently early and specific physiological signature after controlling for alcohol, sleep disruption, motion sickness, exertion, heat, and medication.

Future directions are straightforward. A prospective embarkation cohort could collect symptom histories, travel-group links, molecular stool or rectal-swab results, viral load, genotype, and follow-up illness to estimate the transition from PCR positivity to effective index status. A paired environmental study could measure cabin and bathroom contamination before and after symptom onset to quantify when cleaning and isolation can still prevent onward spread. A pooled-testing pilot could compare individual and pooled molecular results across realistic collection devices and turnaround times. A wearable pilot should be designed as a diagnostic-accuracy study with laboratory confirmation, not as a direct operational escalation trial. Finally, CTB should be rerun with platform-specific parameter files for naval, expedition, and offshore settings rather than applying cruise cost-effectiveness results to those domains.

Sensitivity and decision thresholds

The most sensitive policy conclusion is not the biological effect of screening. It is the threshold at which screening becomes worth buying. Universal PCR screening is effective in the model because it removes introductions before contacts begin. It is not economically preferred in the calibrated cruise case because the effective-index rate is low. If n_{eff} moved from 0.035 to the upper end of the plausible range, if a new genotype raised susceptibility for a season, or if a particular itinerary had repeated exposure to high-risk source populations, the same screening technology would move closer to viability. The framework therefore treats screening as a conditional intervention: a poor default for average cruise voyages, but potentially rational during high-risk windows, for high-consequence pathogens, or when a low-cost gated design enriches the tested population.

The same threshold logic applies to onboard response. Enhanced response is robust in the cruise norovirus analysis because it is triggered after evidence appears. Its weakness is that it cannot stop the first generation of exposure and it depends on timely reporting. If passengers delay reporting or if isolation creates many new crew contacts, effectiveness could fall. If active surveillance increases case finding without increasing unsafe contact workflows, effectiveness could improve. The response parameters that matter most are the time from symptom onset to report, the fraction of symptomatic cases detected, the time from report to isolation, and the number of service contacts created by isolation. These are operational quantities that can be measured on ships and should be prioritized over additional speculative parameter fitting.

Wearables occupy a different threshold space. Their value depends on the time advantage between physiological signal and actionable clinical confirmation. For SARS-CoV-2 or influenza, that interval may be long enough to change contacts before peak transmission. For norovirus, the interval may be too short or too nonspecific, especially if vomiting and environmental contamination occur near symptom onset. A wearable program can still be justified as a low-regret triage layer if it improves situational awareness without triggering high-cost response. It should not be justified as a proven pre-symptomatic norovirus detector. The decision threshold is therefore a combination of specificity, time advantage, and escalation discipline, not simply anomaly sensitivity.

Pooling has a fourth threshold: workflow reliability. The mathematical model can reduce assay counts by orders of magnitude, but the practical system succeeds only if sample collection, barcoding, pool construction, retesting, and reporting occur fast enough to affect boarding or early-voyage decisions. If retesting pushes results into the post-embarkation period, the intervention becomes early onboard surveillance rather than pre-boarding exclusion. That may still have value, but it changes the comparator from universal boarding prevention to enhanced onboard response. The model should therefore report not only assays per voyage and sensitivity by pool size, but also turnaround-time distributions and the fraction of positive pools resolved before operational decisions are made.

These thresholds are why CTB is structured as a counterfactual simulator rather than a single recommendation engine. The correct output is not that one technology wins across all platforms. The correct output is a set of parameter regions: when base rates are low, spend after signal enrichment; when consequence severity is high, tolerate more screening cost; when diagnostics become sub-dollar and fast, revisit pooling; when the evidence base is indirect, keep escalation gated. This interpretation preserves the cruise-specific conclusion while making the platform useful for future vessel types and pathogen profiles.

Validation priorities for implementation

The most useful next validation study would not be a trial of universal screening. It would be a measurement study that estimates the hidden state CTB currently fits: the probability that a PCR-positive boarding traveler is an effective index case. A feasible design would enroll passengers and crew at embarkation during high-risk season, collect a short symptom and exposure history, record travel-group and cabin links, obtain a molecular sample with quantitative Ct or copy-number output, and follow participants for gastrointestinal illness through the voyage and immediate post-disembarkation period. The primary estimand would be the transition from PCR positivity to acute high-titer shedding, symptoms, cabin spread, and reportable medical-center presentation. That design would directly test whether the fitted n_{eff} range is reasonable and would reduce dependence on combining challenge studies, community prevalence studies, and VSP outbreak data.

A second validation priority is environmental timing. Norovirus control depends on when a case contaminates cabins, bathrooms, dining surfaces, and cleaning workflows relative to symptom onset and isolation. The current CTB implementation represents environmental pathways because the literature supports fomite and environmental transmission, but the shipboard time course of contamination is poorly observed. A paired study of symptomatic cases, cabin mates, bathrooms, high-touch surfaces, and cleaning logs could estimate how rapidly contamination rises and how much reduction is achieved by routine versus enhanced disinfection. That evidence would improve estimates for isolation duration, cleaning intensity, food-delivery workflows, and the risk that response activities themselves create additional contacts [10, 25–27].

A third priority is operational validation of diagnostics. For individual tests, the relevant question is not only analytical sensitivity against a reference assay. It is whether the test detects the passengers or crew who are likely to start operationally meaningful chains. That requires quantitative viral-load measurement, sample-type comparison, and time-to-result data under field conditions. For pooled testing, the validation threshold is higher because pooling adds dilution, sample-handling complexity, retesting delays, and sample-tracking risk. A laboratory pool that detects a high-titer sample is not sufficient evidence for an embarkation program. The validation should compare individual and pooled testing on real collection devices, with realistic transport time and staff workflow, and should report false negatives by viral load and shedding phase.

A fourth priority is behavioral response. The model assumes that detection and isolation change contacts in specified ways. In practice, response depends on passenger compliance, crew workload, cabin service, medical-center capacity, language barriers, compensation policies, and the perceived legitimacy of testing. Poorly designed isolation can increase crew contacts or delay self-reporting. A validation program should therefore measure operational outcomes such as time from symptom onset to report, time from report to isolation, number of crew contacts during meal and linen delivery, and the fraction of cabin mates who comply with monitoring. These quantities may be as important as assay sensitivity when the goal is reducing reportable outbreaks.

Interpretation for naval and other non-cruise platforms

The cruise norovirus analysis is the only cost-effectiveness claim made here. The report intentionally avoids assigning costs to naval or military outcomes. That restriction does not limit CTB as a software platform; it limits the inference that should be drawn from the cruise economics. A naval vessel differs from a cruise ship in cohort stability, command authority, berthing density, mission consequence, medical evacuation options, privacy constraints, and tolerance for false alarms. A strategy that fails cruise cost-effectiveness because most voyages have no effective index case could be rational in a mission setting if the consequence of one missed introduction is high enough. Conversely, a strategy that looks inexpensive on a cruise ship could be unacceptable if it creates operational disruption, compromises watch rotations, or generates false-positive isolation during a critical mission period.

For these platforms, CTB should be used by replacing the parameter file rather than reinterpreting the cruise result. The effective-index model still provides a useful structure: distinguish prevalence from transmission competence, distinguish biological efficacy from operational cost, and model response as part of the contact network. The values should be re-estimated. Cohort turnover may be lower, voyages may be longer, and exposure events may be dominated by port calls, maintenance periods, or shore operations rather than embarkation. Crew-first wearable surveillance may also have a different interpretation when there are no passengers and when the monitored population is the whole operational community. These cases are configurable, but they require domain-owned assumptions and validation data.

The same caution applies to offshore platforms, expedition vessels, research cruises, and remote camps. Smaller cohorts and longer deployments can increase the value of repeated testing or wastewa-

ter/environmental monitoring because the same individuals remain at risk over time. Medical evacuation constraints can make early detection more valuable. On the other hand, smaller populations may make stochastic extinction more common and may reduce the value of complex pooling. The model should therefore be read as a framework for counterfactual comparison, not as a universal ranking of interventions.

8 Conclusions

1. **Cruise norovirus outbreaks are rare because effective introductions are rare.** Community PCR positivity is common, but only a small subset of PCR-positive boarders are acute transmission-competent shedders, and overdispersion plus partial immunity extinguish most introductions.
2. **The calibrated CTB reconciliation model fits both baseline rarity and GII.4 emergence.** Rounded simulation parameters of $R_0 = 4.5$, $k \approx 0.10$, susceptibility 0.30, and $n_{\text{eff}} = 0.035$ per voyage reproduce baseline outbreak rarity, while increased susceptibility reproduces emergence-period spikes.
3. **Enhanced onboard response is the best cruise norovirus investment under current assumptions.** It does not prevent introductions, but it reduces cases and reportable outbreaks at low incremental cost.
4. **Universal pre-boarding screening is biologically coherent but economically marginal for cruise ships at current test costs.** Pooled molecular screening becomes plausible only with sub-dollar fully loaded assay costs, seasonal or gated deployment, or a higher-consequence pathogen scenario.

The CTB software is available at [Crusher_to_the_Bridge](#) and is configurable for other confined vessel platforms. Cruise-ship results should not be transferred to naval, offshore, or expedition settings without replacing platform duration, population structure, diagnostic logistics, and cost parameters.

A Supplementary Material: Platform Configuration and Generalizability

This supplementary section lists configuration parameters for applications outside the cruise-norovirus analysis. CTB is configurable for oil rigs, naval deployments, expedition vessels, research cruises, and other confined platforms. The exposed parameters include deployment duration, crew size, passenger or worker population, screening frequency, contact network structure, diagnostic latency, assay cost, isolation capacity, outbreak cost, and reporting thresholds. These parameters are deliberately separate from the cruise calibration so that platform-specific users can replace cruise assumptions rather than inherit them.

The present cost-effectiveness analysis applies to cruise ships. Longer deployments, smaller fixed cohorts, higher disruption costs, and different medical evacuation constraints can change the economics of screening and surveillance. The report does not estimate military operational costs or military outcomes. Naval and other government users would need domain-specific assumptions, threat models, and operational constraints before drawing cost-effectiveness conclusions from CTB.

Table 6: Platform parameters intended to be configured before CTB is applied outside the cruise setting.

Domain	Configurable parameters	Why it changes conclusions
Population	Passenger count, crew count, duty groups, cohort turnover, berth and cabin assignment	Changes contact density, travel-group clustering, crew-sentinel coverage, and the number needed to screen.
Deployment	Voyage duration, port calls, excursions, shore leave, resupply cadence	Changes incubation overlap, exposure opportunities, and whether introduction prevention or onboard response dominates.
Layout	Decks, dining venues, medical spaces, berthing, shared sanitation, HVAC or circulation zones	Changes route-specific transmission, environmental persistence, and isolation feasibility.
Pathogen	R_0 , k , shedding curve, incubation, environmental stability, infectious dose, immunity	Determines whether early detection, cleaning, isolation, or screening is the preferred control point.
Diagnostics	Assay sensitivity, specificity, sample type, pooling design, latency, collection compliance	Determines whether a test detects the operationally relevant state and whether results arrive in time to change contacts.
Response	Isolation duration, food delivery, galley restrictions, cleaning protocols, medical staffing	Can reduce transmission or create new contacts; response must be modeled as part of the contact network.
Economics	Test cost, labor, false-positive cost, outbreak cost, evacuation cost, mission cost	Changes cost-effectiveness without changing biological efficacy.
Reporting	Case definition, outbreak threshold, ascertainment, regulatory trigger	Determines the difference between infections, symptomatic cases, reportable outbreaks, and operational incidents.

For military or naval platforms, the same configuration points are available, but cost and consequence assumptions are not portable from cruise ships. A naval vessel may have smaller cohorts, longer deployments, different berthing density, different command authority, different privacy constraints, and very different loss functions for missed infections or false alarms. CTB can represent these structures, but this report does not assign dollar values to mission disruption, readiness, evacuation, or classified operational outcomes. The correct use is to replace the cruise-specific parameter files and rerun the same counterfactual comparisons under domain-owned assumptions.

Table 7: Pathogen parameters that should be replaced when CTB is configured for non-norovirus use cases.

Parameter	Cruise norovirus role	Generalization requirement
Shedding phase	Separates prolonged PCR positivity from acute transmission competence	Use pathogen-specific viral or bacterial load kinetics and infectiousness evidence.
Dose-response	Links contamination and exposure to infection risk	Replace with pathogen-specific quantitative microbial risk assessment or outbreak-derived estimates.
Transmission routes	Direct, cabin, food-service, fomite, and environmental routes are important	Disable or reweight routes that are biologically implausible for the target pathogen.
Immunity	Partial genotype-specific immunity controls susceptibility and emergence spikes	Configure vaccination, prior exposure, waning, immune escape, and host-genetic susceptibility.
Overdispersion	Low k produces extinction of most introductions and rare large clusters	Estimate or bracket k from pathogen- and setting-specific outbreak data.
Clinical trigger	Gastrointestinal symptoms drive reporting and isolation	Replace with syndrome-specific reporting, severity, and care-seeking behavior.

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